11.6 Theoretical Distributions: Some Special Discrete Distributions

In the previous section we have dealt with various general probability distributions with mean and variance. We shall now learn some discrete probability distributions of special importance.

In this section we learn the following discrete distributions.

(i) The One point distribution
(ii) The Two point distribution
(iii) The Bernoulli distribution
(iv) The Binomial distribution.

11.6.1 The One point distribution

The random variable X has a one point distribution if there exists a point x_0 such that, the probability mass function $f(x)$ is defined as $f(x) = P(X = x_0) = 1$.

That is the probability mass is concentrated at one point.

The cumulative distribution function is

$$F(x) = \begin{cases} 0 & -\infty < x < x_0 \\ 1 & x_0 \le x < \infty \end{cases}$$

Mean :

$$E(X) = \sum_x x\, f(x) = x_0 \times 1 = x_0$$

Variance :

$$V(X) = E(X^2) - \left(E(X)\right)^2 = \sum_x x^2 f(x) - \left(x_0\right)^2 = {x_0}^2 - {x_0}^2 = 0$$

Therefore the mean and the variance are respectively x_0 and 0.

11.6.2 The Two point distribution

(a) Unsymmetrical Case: The random variable X has a two point distribution if there exists two values x_1 and x_2, such that

$$f(x) = \begin{cases} p & \text{for } x = x_1 \\ 1-p & \text{for } x = x_2 \end{cases} \qquad \text{where } 0 < p < 1.$$

The cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{if } x < x_1 \\ p & \text{if } x_1 \le x < x_2 \\ 1 & \text{if } x \ge x_2 \end{cases}$$

Mean :

$$E(X) = \sum_x x\, f(x) = x_1 \times p + x_2 \times (1-p) = px_1 + qx_2 \text{ where } q = 1-p.$$

Variance :

$$V(X) = E(X^2) - \left(E(X)\right)^2 = \sum_x x^2 f(x) - \left(px_1 + qx_2\right)^2$$

$$= \left({x_1}^2 p + {x_2}^2 q\right) - \left(px_1 + qx_2\right)^2 = pq\left(x_2 - x_1\right)^2$$

The mean and the variance are respectively $px_1 + qx_2$ and $pq\left(x_2 - x_1\right)^2$

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